



C PROGRAM PRACTICE

Bangalore

TITBITS

1. “? : ” is the only ternary operator in C language.
2. “sizeof” is the only operator which is also a keyword.
3. In printf() and scanf(), f stands for formatted not function.
4. Compiler doesn't know header file, preprocessor processes these and expands source code.
5. Header file name can be included in 2 ways, (a) Using angular brackets (b) Using double inverted comma.
6. Compiler generates assembly code not machine code and then assembler generates the machine code.
7. sizeof operator cannot tell us the size of functions because this operator works on compile time and functions loads in memory at run time.

COMPILATION

1. C compilation process in detail
2. What is segmentation fault?
3. What are the common causes of segmentation fault in C

ARRAYS

1. Program to find the largest element in an array
2. Find second largest number in an array
3. Program to reverse a string
4. Addition, Deletion and swapping elements in an array.
5. consider in an array having 5 elements 1,2,3,4,5 from this one of them is missing, suppose 4 is missing... how you will find in efficient way?
6. Write a program to return the number of words either start or end with vowels in a string
7. Explain what p holds `int (**p)[10];`//it is pointer to an array pointer
8. Explain what p hold `int (*p)[10];`//this will hold the address of first element in the array
9. Check anagram of string
10. There are two constant arrays, A and B. Multiply A and B and store it in the array C
11. Reverse a string with fixed character array and without fixed array size
12. How do you dynamically allocate memory in an array
13. Write a C program to transpose a given matrix. The transpose of a matrix is obtained by swapping its rows with columns.
14. Write a C program to multiply two matrices of compatible dimensions using pointers. Display the resulting matrix.
15. Write a C program to find the sum of diagonal elements in a square matrix.

1. Write a C program to sort the rows of a matrix in ascending order based on the first element of each row.
2. Write a C program to find the maximum element in each row of a matrix and store the results in a one-dimensional array.
3. Write a C program to find the trace of a square matrix (sum of diagonal elements).
4. Write a C program to calculate the determinant of a 2x2 matrix using pointers.
5. Write a C program to check whether a given matrix is symmetric or not.
6. Write a C program to find the inverse of a 2x2 matrix using pointers.
7. Write a C program to represent a sparse matrix using a two-dimensional array and display the original and sparse matrix.

8. `Int arr[][]={{1,2 3 4 5},{6,7,8,9,10},{11,12,13,14,15},{16,17,18,19,20}};`

`int *per = arr+3;`

`Print ("%d",*per[1])`

`int *per = *arr+3`

`Print ("%d",*per[1])`

9. `str = "Hello, welcome to the welcomeworld of C programmingwelcome"`

Sub string : welcome

Find indexes from where sub string started

10. `int arr[] = {1,2,2,3,4,4,4,5,2,6,7}`

Frequency of occurrence of each data

STRUCTURE

1. Determine structure size with and without padding
2. what is the use of structure padding?
3. Configure SPI clock frequency register using structure
4. Define a structure named Circle to represent a circle with a radius. Write a C program to calculate the area and perimeter of two circles and display the results.
5. Design a structure named "Car" to store details like car ID, model, and rental rate per day. Write a C program to input data for three cars, calculate the total rental cost for a specified number of days, and display the results.
6. Write a program to add, subtract and multiply two complex numbers using structures to function.
7. Write a structure to store the name, account number and balance of customers (more than 10) and store their information.
 - 1 - Write a function to print the names of all the customers having balance less than \$200.
 - 2 - Write a function to add \$100 in the balance of all the customers having more than \$1000 in their balance and then print the incremented value of their balance.
8. How to access structure member using structure address
9. Is it possible to use like this, `uint *ptr = &my_struct`
10. Difference between enum and structure ?
11. Declare a structure having pointer, integer, array, void pointer, pass it in function and initialise it.
12. Declare a structure with `int`, `char[]` and boolean data type and initialise the structure in compile time and print the values

STRUCTURE(CONTD.)

for ex,

```
struct spi
{
    field_1:4,
    field_2:4,
    field_3:4,
    field_4:4,
}spi_config;
```

how the above structure will reflect in h/w? How the i/o mapping is happening? How s/w configuration reflection in h/w?

why do we need structure in C?

declare the structure having pointer,integer,array,void pointer,pass it in function and initialise it.

DATA STRUCTURE

1. Implement a linked List to store employee details viz Name, Emp ID, Age, Dept name. It should provide to addition and deletion of employee record.
2. Write a C program to remove duplicates from a single unsorted linked list
3. Remove the middle node from the linked list
4. how to find whether the linked list is cyclic
5. Sort the array in ascending order using bubble sortWrite a C program to detect and remove a loop in a singly linked list.

Test Data and Expected Output :

Original singly linked list:

1 2 3 4 5

Create the loop:

Following statement display the loop:

displayList(head);

After removing the said loop:

1 2 3 4 5

VARIABLES

1. Difference between Extern and Static variable.
2. If need to share static variable value(which is declared as a global) from one file to another file, then what I need to do
3. What is Typedef, Typecasting, Static, volatile uses
4. Draw the memory layout, and tell me where the memory allocated for the variables
5. What happen if we define static variable in header file and call in 2 different c file
6. How can we read or write a value to fixed memory location(write a program)
7. Difference between Const and volatile
8. Program to reverse a number
9. Write a program and give a value and read it, increment that value by 1 and store that incremented value into same location
10. What is the use of extern keyword
11. what is the use of static keyword
12. what is the use of typedef
13. What will happen if we declare a static variable outside a function?

VARIABLE(CONTD.)

1. Explain pre & post increment.
2. Write a program to convert string "1234" to decimal 1234
3. Write a C program to read the roll no, name and marks of three subjects and calculate the total, percentage and division.

Test Data :

Input the Roll Number of the student :784

Input the Name of the Student :James

Input the marks of Physics, Chemistry and Computer Application : 70 80 90

Expected Output :

Roll No : 784

Name of Student : James

Marks in Physics : 70

Marks in Chemistry : 80

Marks in Computer Application : 90

Total Marks = 240

Percentage = 80.00

Division = First

FUNCTIONS

1. Write a C program to find whether the number provided is Armstrong or not. It will have two functions viz Func1() to get the input from the user & Func2() to check if the input number is Armstrong. Also, need to implement exceptional handling so that programs throw an error if the number provided is non integer and prompt user to enter integer value.
2. Recursion function (What happens if the base condition is not defined in recursion)
3. Write a program to get the factorial of a number by using recursive functions
4. Write a program in function as integer variable and return byte by byte.
5. There are two files and having a function with same name with or without static in two files. Explain this scenario.
6. Write your own memcpy()
7. Write a function to find the loop in the linked list.
8. Write a program to reset a particular bit.
9. Implement memmov function
10. What is the difference between Static and normal function.
11. How Macro is different than function
12. What is Inline function
13. write function to find the given number is prime or not.
14. Program to check if the system is big or little endian?
15. what is re - entrant function?
16. Write a program to transfer message between two function

FUNCTIONS(CONTD.)

1. Largest element and number of times an element is repeated in a single loop iteration
2. We have 2 functions in a c file
 - function 1 : formatting flash device, updating global var : % update info
 - function 2 : printing the flash percentage every 1 sec.
 - Make function 3 to print the Global variables to update status of the flashing update percentage.

FUNCTIONS(CONTD.)

1. Write a Macro function to clear bit
2. display number 0-10 using recursive function.
3. Write a macro to toggle mth bit with m no of times?
4. How to change the start the function before executing main function?
5.

```
. int fun(int **arr, int * col, int * row)
{
    // Fill code with all error checking condition and intialize with zero
```
6. copy source to destination through the pointer

```
void copy(int *d,int *s,int n)
{
    int i;
    for(i=0;i<n;i++)
        *d++=*s++;
}
```

error possibilities in above code? \

POINTERS

1. What is the difference between
 `Char *strg1 = "Mirafra"`
 `Char strg2[] = "Mirafra"`
2. Write a program in C to demonstrate the use of pointers to structures
3. Write a C program to return multiple value from function using pointers.
4. Write a C program to add two numbers using a function pointer.
5. Write a C program to sort an array using a function pointer
6. Write a program in C to implement a callback function using a pointer
7. Write a program in C to reverse a string using a pointer
8. Write a C program to print the square of array elements using callback function
9. What is Dangling pointer
10. Explain `const int*ptr` and `int const*ptr`
11. Write a C code which uses function pointer in it
12. What is wild pointer
13. how to access the `void *value` using `ptr`.
14. what is the difference between function pointer and normal pointer?
15. why we need function pointer?
16. what is double pointer? what is the use of double pointer?
17. what is the difference between double pointer and single pointer?
18. what is void pointer?
19. what is dangling pointer? Give me some example?

POINTER

```
1.  short a=320;

    unsigned char * ptr=NULL;

    ptr=&a;

    printf("%d",*ptr);
```

```
2.  int arr[]={ 1,2,3,4,5,6};
```

```
int *p=arr;

++*p;

p+=2;

printf("%d",*p);
```

3. what is double pointer?what is the use of double pointer?

4. what is the difference between double pointer and single pointer?

5. int main()

```
{

    int a,b,c;

    char *p=0;

    int *q=0;

    double *r=0;

    a=(int)(p+1);

    b=(int)(q+1);

    c=(int)(r+1);
```

POINTER

1. what is dangling pointer?give me some example?

1. Write an endianness program input : 0x12345678 , output : 0x78563412
2. Write a program to check if the system is big or little endian?
3. How to store 32 bit data into a 16 bit MCU (write a program for it).
4. Swap all even and odd bits
5. Write a program to achieve the output for the given input. Input data = 0x1234; output data = 0x4321
6. Find the number of bits in a number
7. Swap nibbles in 1 byte number.
8. Toggle and clear bit in given number
9. write a c program to find non repeating element in an array without using loop and by using bitwise operator
10. Write a C program to find the no. of 1's in a given unsigned integer.
11. Write a C program to find endianness of a system
12. Write program to conversion of decimal to hexadecimal
13. What is Access register and set a particular bit
14. Write a program where 0xAB and 0xCD are stored in different integer variables; perform bitwise operators such that the expected output is 0xBC.
15. Write a program to reverse the hex value using bitwise
16. they given one 8bit register in that we have to change last 3 bit without change other bit

MEMORY

1. How can we read or write a value to fixed memory location(write a program)
2. Print even and odd no's using two threads, so that at a time only one thread can access the common resource.
3. Write a C program to check a stack is full or not using an array with push and pop operations.
4. Write a C program to find the minimum element in a stack

Current stack elements:

9 2 4 2 4

Minimum element: 2

5. Write C code by considering the memory address 0x6000_0000, read the value from this address, increment value by 1 and store the value back to same address.
6. Describe the Memory layout of C.
7. Two functions(fun_a & fun_b) having same variable as Static int a ,now we calling that two functions how the memory will allocate for the variable a.
8. What is Bss segment in C

MEMORY(CONTD.)

1. How -1 is storing in memory
2. What is MMU
3. What is virtual memory? why do we use it?
4. Which part of memory is a string variable stored?
5. Given a range location address 1500 to 2500. Shift the data to 2000 to 3000 location

ALGORITHM

1. Write a C program that checks if a positive integer is divisible by either 3 or 7, or both. If the integer is a multiple of 3, then the program will return true. Similarly, if the integer is a multiple of 7, then also the program will return true. If the integer is not a multiple of 3 or 7, then the program will return false.
2. Write a C program that checks two given temperatures and returns true if one temperature is less than 0 and the other is greater than 100, otherwise it returns false.
3. Write a C program to check two given integers whether either of them is in the range 100..200 inclusive.
4. Write a C program to check which number is nearest to the value 100 among two given integers. Return 0 if the two numbers are equal.
5. Write a C program to check if two given non-negative integers have the same last digit.
6. Write a C program that checks if a given non-negative integer is a multiple of 13 or one more than a multiple of 13. For example, if the given integer is 26 or 27, the program will return true, but if it is 25 or 28, the program will return false.
7. Write a C program to check two given integers. Each integer is in the range 10..99. Return true if a digit appears in both numbers, such as the 3 in 13 and 33

STORAGE CLASS

1. Difference between `const int* ptr;` and `int* const ptr.`
2. Give the example for `const int* ptr;` and `int* const ptr.`
3. Same static variable is initialize in 2 different file then what happens?
4. Syntax of constant pointer to constant
5. How to access particular address with volatile keyword write syntax

WHAT IS THE OUTPUT

```
typedef enum {  
    INT,  
    FLOAT,  
    POINTER  
}typeName;  
  
typedef struct {  
    typeName type;  
    void *value;  
}Value;
```


WHAT IS THE OUTPUT

```
#include <stdio.h>

int main()
{
    int array[5];
    int *ptr=&array[0];
    printf("%d", *(ptr+7));
    return 0;
}
```

WHAT IS THE OUTPUT?

```
#include <stdio.h>

int main()
{
    int i;
    for (i = 0; i <= 5; i++) ;
    {
        printf("MiraFra Technologies\n");
    }
    return 0;
}
```

WHAT IS THE VALUE?

1. what is the value of var

```
union test{  
    int a;  
    int b;  
    int c;  
    float f;  
};  
t1.c=2;  
t1.a=3;  
t1.b=5;  
t1.b= t1.a * t1.b * (t1.c++)  
var = t1.c;  
3.
```

2. what is the value of f?

```
union test{  
    int a;  
    int b;  
    char c;  
    float f;  
}t1;  
int t1.c=10
```

FIND THE 2ND LARGEST ELEMENT IN AN ARRAY WITHOUT SORTING.

```
#include <stdio.h>
#include<limits.h>

int find_second(int arr[],int size){
    if(size<2){
        printf("array must have at least two elements\n");
        return 0;    }
    int first=INT_MIN,second=INT_MIN;
    for(int i=0;i<size;i++){
        if(arr[i]>first){            second=first;
            first=arr[i];        }
        else if(arr[i]>second && arr[i]<first){
            second=arr[i];        }    }    return second; }

int main() {
    int arr[]={12,35,1,10,34,1};
    int size=sizeof(arr)/sizeof(arr[0]);
    int secondlargest=find_second(arr,size);
    printf("The second largest element is %d\n",secondlargest);    return 0; }
```

WHAT IS THE OUTPUT?

```
#include <stdio.h>
int main()
{
    int array[5];
    int *ptr=&array[0];
    printf("%d", *(ptr+7));
    return 0;
}
```

WHAT IS THE OUTPUT

1. what is the size of union

```
union test{  
    int a;  
    int b;  
    char c;  
    float f;  
};
```

2. what is the size of structure

```
union test{  
    int a;  
    int b;  
    char c;  
    float f;  
};
```

3. union test t1,t2;

```
struct test_struct{  
    union t3;  
    union t4;  
};
```

WHAT IS THE OUTPUT?

```
int x = 3;

switch case(x)

{

case 1: printf("A");

case 2: printf("B");

case 3: printf("C");

case 4: printf("D");

case 5: printf("E");

default:

printf("F");

}
```

WHAT IS THE OUTPUT?

```
#define MACRO(X) X + 100

int main()

{
    int iVarient = 7;
    printf("==> %d", 10 * MACRO(++iVarient));
    return 0;
}
```


WHAT IS THE OUTPUT?

```
#include<stdio.h>

int main()
{   int a, b, c;
    char *p = 0;
    int *q = 0;
    double *r = 0;
    a = (int)(p + 1);
    b = (int)(q + 1);
    c = (int)(r + 1);
    printf("%d %d  %d",a, b, c);
    return 0; }
```

WHAT IS THE OUTPUT?

```
Main()
{
    Char Str1[] = "Mirafra";
    Char str2[] = "Technologies";
    strcpy(str1, str2);
    Printf("%s",str1);
}
```

WHAT IS THE OUTPUT?

what will be the Expected output and why

```
char a=255;
```

```
a++;
```

```
printf("%d",a);
```

WHAT IS THE OUTPUT?

what will be the Expected output and why

```
char a=255;
```

```
a++;
```

```
printf("%d",a);
```

WHAT IS THE OUTPUT?

what will be the expected output

```
int a=255;
```

```
printf("%d",++a++)
```

WHAT IS THE OUTPUT?

```
uint8_t i = 250;  
for (; i < 255; i = i+2)  
{  
    printf("i = %d\n", i);  
}
```

WHAT IS THE ANSWER?

```
#include <stdlib.h>
```

```
int *f1()
```

```
{ int x = 10; return &x; }
```

```
int *f2()
```

```
{ int *ptr; *ptr = 10; return ptr; }
```

```
int *f3()
```

```
{ int *ptr; ptr = (int*) malloc(sizeof (*ptr)); return ptr;  
}
```

Which of these functions uses pointers incorrectly?

- (a) f3 only
- (b) f1 and f3
- (c) f1 and f2
- (d) f1, f2, and f3

WHAT IS THE OUTPUT?

```
#include <stdio.h>

const volatile unsigned int local = 10;

int main(void)
{
    int *ptr = (int*) &local;

    printf("Initial value of local : %d \n", local);

    *ptr = 100;

    printf("Modified value of local: %d \n", local);

    return 0;
}
```


WHAT IS THE OUTPUT?

```
Int *Ptr = (int *)malloc(100 * sizeof(int));
```

```
Ptr++;
```

```
Ptr++;
```

```
Ptr++;
```

```
Ptr++;
```

```
Ptr++;
```

```
Free(Ptr);
```

WHAT IS THE OUTPUT?

```
#define MACRO(X) X + 100

int main() {
    int iVarient = 7;
    printf("==> %d", 10 * MACRO(++iVarient));
    return 0;
}
```

WHAT IS THE ANSWER?

```
Char *str1 = "Qualcomm"
```

```
Char str2[] = "Qualcomm"
```

What is the difference between Str1 and Str2?

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
    int a,b,c;
```

```
    char *p=0;
```

```
    int *q=0;
```

```
    double *r=0;
```

```
    a=(int)(p+1);
```

```
    b=(int)(q+1);
```

```
    c=(int)(r+1);
```

```
    printf("%d %d %d",a,b,c);
```

```
}
```

what is the value of a,b,c.

WHAT IS THE OUTPUT?

```
int x = 3;

switch case(x)
{
case 1: printf("A");
case 2: printf("B");
case 3: printf("C");
case 4: printf("D");
case 5: printf("E");
default:
printf("F");
}
```

WHAT IS THE OUTPUT?

what will be output?

```
int main () {  
    main ();  
}
```

HOW TO?

How to assign "MIRAFRA TECHNOLOGIES" to ptr through ptest ?

```
struct test
{
    char *ptr;
};

int main()
{
    struct test *ptest;
}
```

HOW TO?

How to print “void pointer” as an output

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
char *ptr = "void pointer";
```

```
void *vptr;
```

```
vptr = &ptr;
```

```
printf("%s" , ?);
```

```
return 0;
```

```
}
```

ADDITIONAL

1. What is thread? How it's created?
2. Have you used any debugging tools(if yes, which debugger you have used and how can we check the local variable in watch window)
3. When does stack overflow
4. How does compiler work(what is meant by 16/32 bit compiler)
5. Code/Algorithm optimization factors that are to be considered
6. Consider 1000 samples, as a developer what are the factors considered to select the best among the 1000 samples.
7. What are the factors or requirements for improving a firmware code such that it increases the performance
8. What is Deadlock and Memory leakage
9. What IDE you have used ?
10. What version control you have used ?
11. What process you followed for your development ?
12. Have you used Keil IDE for development ?
13. How did you debugged your code ?

ADDITIONAL(CONTD.)

1. Can we flash a code compiled in 16 bit compiler on a 32 bit target board.
2. How to store 32 bit data into a 16 bit MCU (write a program for it). .
3. what is Stack. How does data gets stored in it. Are there any exceptions.
4. How many Linker registers are there?
5. What are the controllers you are familiar with ?
6. write the program for

```

*
* *
* * *
* * * *
* * *
* * *
* *
*
```
7. What is MISRA C Violations

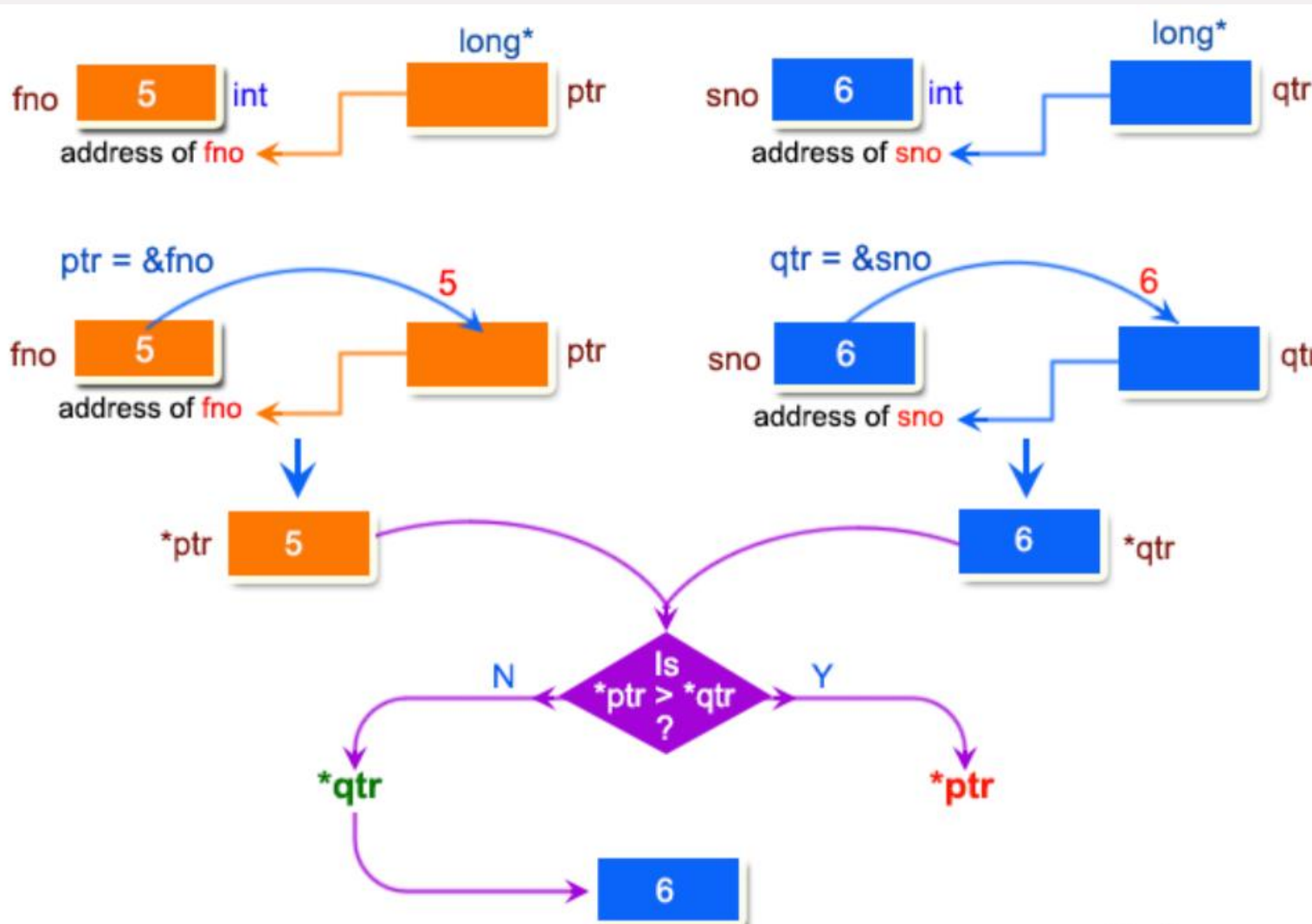
ADDITIONAL(CONTD.)

1. Add two numbers without using the addition operator.
2. What should be the if condition in the following code snippet to print 'Hello World'?

```
if "condition"  
    printf ("Hello");  
Else  
    printf("World");
```
3. Swap 2 numbers without using a third variable.
4. Print a semicolon without using a semicolon in the program
5. Print numbers from 1 to N without using any looping construct.
6. Determine if 2 integers are equal without using comparison and arithmetic operators
7. Given 2 numbers. Find the minimum number without using conditional statements or ternary operators.
8. Consider in an array having 5 elements 1,2,3,4,5 from this one of them is missing, suppose 4 is missing... how you will find in efficient way?

REFERENCE PROGRAM

Write a program in C to find the maximum number between two numbers using a pointer.



C Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main() {
5      int fno, sno, *ptr1 = &fno, *ptr2 = &sno;
6
7      printf("\n\n Pointer : Find the maximum number between two numbers :\n");
8      printf("-----\n");
9
10     printf(" Input the first number : ");
11     scanf("%d", ptr1); // Read the first number from the user and store it using ptr1
12     printf(" Input the second number : ");
13     scanf("%d", ptr2); // Read the second number from the user and store it using ptr2
14
15     // Compare the values pointed by ptr1 and ptr2 to find the maximum number
16     if (*ptr1 > *ptr2) {
17         printf("\n\n %d is the maximum number.\n\n", *ptr1); // Print the maximum number
18     } else {
19         printf("\n\n %d is the maximum number.\n\n", *ptr2); // Print the maximum number
20     }
21     return 0;
22 }
```

REFERENCE PROGRAM

- Swapping of bits. User can give the bit positions to be swapped as an input. Here the number provided is 0xAA.

```
#include <stdio.h>
```

```
void swap_two_bits(int num, int p1, int p2) {
```

```
    unsigned int mask = 1, mask1, mask2;
```

```
    mask1 = mask << (p1-1);    mask2 = mask << (p2-1);
```

```
    mask = mask1 | mask2;
```

```
    num ^= mask;
```

```
    printf("Value after swap two bits is %x\n", num); }
```

```
int main()
```

```
{    int num=0xAA, p1, p2;
```

```
    printf("Enter position that youu want to swap\n");
```

```
    scanf("%d %d", &p1, &p2);
```

```
    swap_two_bits(num, p1, p2);
```

```
    return 0;
```

```
}
```